

REVTeX4

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Abstract

Shell document for REVTeX 4.

Contents

A. (**Real**) $\dim \{V_2\} = 2 \cong \mathbb{R}^2$

a. Basic Properties A complex unit in V_2 is given by

$$J_e = |e_2\rangle \langle e_1| - |e_1\rangle \langle e_2| \quad (1)$$

$$R_e(J_e) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (2)$$

the associated complex conjugation

$$\sigma_e : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad (3)$$

$$\sigma_e = |e_1\rangle \langle e_1| - |e_2\rangle \langle e_2| = |e_1\rangle \langle e_1| - J_e |e_1\rangle \langle e_1| J_e^t \quad (4)$$

$$R_e(\sigma_e) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (5)$$

and the definition of real

$$|a_{1r}\rangle = \frac{1}{2} (|a_1\rangle + \sigma_e(|a_1\rangle)) = J_e(|e_1\rangle) a_{11} \quad (6)$$

$$R_e(|a_{1r}\rangle) = \frac{1}{2} \left(\begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} + \begin{pmatrix} a_{11} \\ -a_{12} \end{pmatrix} \right) = \begin{pmatrix} a_{11} \\ 0 \end{pmatrix} \quad (7)$$

and complex parts

$$|a_{1i}\rangle = \frac{1}{2} J_e (|a_1\rangle - \sigma_e(|a_1\rangle)) = J_e(|e_1\rangle) a_{12} \quad (8)$$

$$R_e(|a_{1i}\rangle) = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \left(\begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} + \begin{pmatrix} a_{11} \\ -a_{12} \end{pmatrix} \right) \quad (9)$$

$$= \frac{1}{2} \left(\begin{pmatrix} -a_{12} \\ a_{11} \end{pmatrix} + \begin{pmatrix} a_{12} \\ a_{11} \end{pmatrix} \right) = \begin{pmatrix} 0 \\ a_{12} \end{pmatrix} \quad (10)$$

A representation of $\mathbb{C}$ in V_2 is given by

$$K_e(z_1) = (a_{11}I_2 + a_{12}J_e); a_{11}, a_{12} \in \mathbb{R}. \quad (11)$$

b. Complexification Maps The inverse vector complexification map is a real linear map

$$\mathfrak{C}_e^{-1} : V_{2\mathbb{C}} \rightarrow V_2 \quad (12)$$

$$\mathfrak{C}_e^{-1}(|e_1\rangle) = |e_1\rangle \quad (13)$$

$$\mathfrak{C}_e^{-1}(i|e_1\rangle) = J_e|e_1\rangle = |e_2\rangle \quad (14)$$

and the inverse operator complexification map

$$\widehat{\mathfrak{C}}_e^{-1}(X) = \mathfrak{C}_e^{-1}X\mathfrak{C}_e \quad (15)$$

$$\widehat{\mathfrak{C}}_e^{-1}(I_{V_{2\mathbb{C}}}) = I_{V_2} = |e_1\rangle\langle e_1| + |e_2\rangle\langle e_2| \quad (16)$$

$$\widehat{\mathfrak{C}}_e^{-1}(iI_{V_{2\mathbb{C}}}) = \mathfrak{C}_e^{-1}(i|e_1\rangle\langle e_1|) = J_e|e_2\rangle\langle e_1| - |e_1\rangle\langle e_2| \quad (17)$$

gives

$$\widehat{\mathfrak{C}}_e^{-1}(zI_{V_{2\mathbb{C}}}) = \widehat{\mathfrak{C}}_e^{-1}(a_{1r}I_{V_{2\mathbb{C}}} + a_{1i}iI_{V_{2\mathbb{C}}}) = a_{1r}I_{V_2} + a_{1i}J_e \quad (18)$$

$$R_e(a_{1r}I_{V_2} + a_{1i}J_e) = \begin{pmatrix} a_{1r} & -a_{1i} \\ a_{1i} & a_{1r} \end{pmatrix}. \quad (19)$$

The operator complexification

$$\widehat{\mathfrak{C}}_e(X) = \mathfrak{C}_eX\mathfrak{C}_e^{-1} \quad (20)$$

is given by

$$\widehat{\mathfrak{C}}_e(I_{V_2}) = \mathfrak{C}_e(|e_1\rangle\langle e_1| + |e_2\rangle\langle e_2|) = I_{V_{2\mathbb{C}}} \quad (21)$$

$$\widehat{\mathfrak{C}}_e(J_e) = \mathfrak{C}_e(|e_2\rangle\langle e_1| - |e_1\rangle\langle e_2|) = iI_{V_{2\mathbb{C}}} = i|e_1\rangle\langle e_1|. \quad (22)$$

The Hilbert space $V_{2\mathbb{C}}$ has an associated real bilinear form that is independent of the basis of $V_{2\mathbb{C}}$

$$(\cdot|\cdot)_{V_{2\mathbb{C}}} : V_{2\mathbb{C}} \times V_{2\mathbb{C}} \rightarrow \mathbb{R} \quad (23)$$

$$(z_1|z_2)_{V_{2\mathbb{C}}} = \operatorname{Re} \langle z_1|z_2 \rangle \quad (24)$$

with the properties

$$(e_1|ie_1)_{V_{2\mathbb{C}}} = 0 \quad (25)$$

$$(e_1|e_1)_{V_{2\mathbb{C}}} = 1 \quad (26)$$

$$(ie_1|ie_1)_{V_{2\mathbb{C}}} = 1 \quad (27)$$

and is non degenerate i.e.

$$(z|z)_{V_{2\mathbb{C}}} = 0 \Leftrightarrow |z\rangle = 0. \quad (28)$$

The form $(\cdot|\cdot)_{V_{2\mathbb{C}}}$ induces a real linear functional, $(z|_{V_{2\mathbb{C}}}$, on $V_{2\mathbb{C}}$ defined by

$$(z|_{V_{2\mathbb{C}}} : V_{2\mathbb{C}} \rightarrow \mathbb{R} \quad (29)$$

$$(z|_{V_{2\mathbb{C}}}(|\omega\rangle) = (z||\omega\rangle)_{V_{2\mathbb{C}}} = \text{Re} \langle z|\omega\rangle \quad (30)$$

i.e.

$$(z|_{V_{2\mathbb{C}}}(\alpha_1|\omega_1\rangle + \alpha_2|\omega_2\rangle) = \alpha_1(z|_{V_{2\mathbb{C}}}(|\omega_1\rangle) + \alpha_2(z|_{V_{2\mathbb{C}}}(|\omega_2\rangle)); \alpha_1, \alpha_2 \in \mathbb{R} \quad (31)$$

$$(z|_{V_{2\mathbb{C}}}(\alpha_1|\omega_1\rangle + \alpha_2|\omega_2\rangle) \neq \alpha_1(z|_{V_{2\mathbb{C}}}(|\omega_1\rangle) + \alpha_2(z|_{V_{2\mathbb{C}}}(|\omega_2\rangle)); \alpha_1, \alpha_2 \in \mathbb{C}. \quad (32)$$

Theorem 1

$$T : |z\rangle \rightarrow (z|_{V_{2\mathbb{C}}} \quad (33)$$

is injective i.e.

$$T(|z_1\rangle - |z_2\rangle) = 0 \Leftrightarrow |z_1\rangle = |z_2\rangle \quad (34)$$

Proof.

$$(z|z)_{V_{2\mathbb{C}}} = \text{Re} \langle z|z\rangle = \langle z|z\rangle \quad (35)$$

thus if

$$((z_1 - z_2)|z) = 0 \forall |z_2\rangle \in V_{2\mathbb{C}} \quad (36)$$

then in particular

$$((z_1 - z_2)|(z_1 - z_2)) = \langle (z_1 - z_2)|(z_1 - z_2)\rangle = 0 \Rightarrow z_1 = z_2 \quad (37)$$

and if

$$z_1 = z_2 \quad (38)$$

then

$$((z_1 - z_2)|z) = 0 \forall |z_2\rangle \in V_{2\mathbb{C}}. \quad (39)$$

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c. Dual Space Representation A complexification map and its inverse can then be expressed as the real linear maps

$$\mathfrak{C}_e = |e_1\rangle (e_1| + |ie_1\rangle (e_2| = (|e_1\rangle, |ie_1\rangle) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} (e_1| \\ (e_2| \end{pmatrix} \quad (40)$$

and

$$\mathfrak{C}_e^{-1} = |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_2\rangle (ie_1|_{V_{2\mathbb{C}}} = |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + J_e |e_1\rangle (ie_1|_{V_{2\mathbb{C}}} = (|e_1\rangle, |e_2\rangle) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} (e_1|_{V_{2\mathbb{C}}} \\ (ie_1|_{V_{2\mathbb{C}}} \end{pmatrix}, \quad (41)$$

which give with respect to the real structure of $V_{2\mathbb{C}}$

$$\mathfrak{C}_e |e_1, e_2\rangle = |\mathfrak{C}_e(e_1), \mathfrak{C}_e(e_2)\rangle = |e_1, ie_1\rangle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = |e_1, ie_1\rangle R_e(\mathfrak{C}_e) \quad (42)$$

$$\mathfrak{C}_e^{-1} |e_1, ie_1\rangle = |\mathfrak{C}_e^{-1}(e_1), \mathfrak{C}_e^{-1}(ie_1)\rangle = |e_1, e_2\rangle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = |e_1, e_2\rangle R_e(\mathfrak{C}_e^{-1}). \quad (43)$$

This leads to

$$\mathfrak{C}_e \mathfrak{C}_e^{-1} = (|e_1\rangle (e_1| + |ie_1\rangle (e_2|) (|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_2\rangle (ie_1|_{V_{2\mathbb{C}}})) \quad (44)$$

$$= |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + 0 + 0 + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}} = |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}} = I_{V_{2\mathbb{C}}} \quad (45)$$

and

$$\mathfrak{C}_e^{-1} \mathfrak{C}_e = (|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_2\rangle (ie_1|_{V_{2\mathbb{C}}})) (|e_1\rangle (e_1| + |ie_1\rangle (e_2|) = |e_1\rangle (e_1| + |e_2\rangle (e_2| = I_{V_2}. \quad (46)$$

The complex unit, $J_{e\mathbb{C}}$, (which is real linear) associated with a basis $\{\mathbf{e}\}$ can be represented in $V_{2\mathbb{C}}$ by

$$J_{e\mathbb{C}} = |ie_1\rangle (e_1|_{V_{2\mathbb{C}}} - |e_1\rangle (ie_1|_{V_{2\mathbb{C}}} \quad (47)$$

$$J_{e\mathbb{C}} |z_1\rangle = (a_{11} + ia_{12}) |e_1\rangle = i(a_{11} + ia_{12}) |e_1\rangle = i |z_1\rangle. \quad (48)$$

The complex conjugation map, $\sigma_{e\mathbb{C}}$, (which is real linear) associated with this basis can be expressed as

$$\sigma_{e\mathbb{C}} = |e_1\rangle (e_1|_{V_{2\mathbb{C}}} - |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}}$$

and thus the definition of $\mathbf{e}$ -real vectors becomes

$$\sigma_{e\mathbb{C}}(|z\rangle) = |z\rangle. \quad (49)$$

The operator

$$|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}}) \quad (50)$$

resolves vectors $|z_1\rangle \in V_{2\mathbb{C}}$ into their real and complex parts wrt the basis $\{|e_1\rangle\}$ i.e.

$$(|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}})) |z_1\rangle = |e_1\rangle (e_1| z_1\rangle) + |ie_1\rangle (ie_1| z_1\rangle) = |e_1\rangle a_{11} + |ie_1\rangle a_{12} = |e_1, ie_1\rangle R_e(a_1) \equiv |e_1, ie_1\rangle \quad (51)$$

B. Detailed Example of Complexification

d. (*Real*) $\dim \{V_2\} = 2 \cong \mathbb{R}^2$ In order to understand a complexification, $V_{2s\mathbb{C}}$, of a real even dimensional vector space V_{2s} , we first consider

$$(\text{real}) \dim \{V_2\} = 2; (\text{complex}) \dim \{V_{2\mathbb{C}}\} = 1, \quad (52)$$

where

$$|e_1\rangle = |\mathbf{e}\rangle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (53)$$

$$|e_2\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (54)$$

and the complexification, J_e , is given by

$$J_e |e_1\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |e_2\rangle \quad (55)$$

$$J_e |e_2\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\mathbf{e}\rangle \begin{pmatrix} -1 \\ 0 \end{pmatrix} = -|e_1\rangle. \quad (56)$$

Hence

$$J_e = |e_2\rangle (e_1| - |e_1\rangle (e_2|), \quad (57)$$

so corresponds to a real antisymmetric bilinear form, g ,

$$g(a, b) = (g| a \wedge b) = (a| J_e b) \quad (58)$$

$$|g\rangle = |e_1 e_2\rangle \quad (59)$$

on V_2 , while

$$I_2 = |e_1\rangle \langle e_1| + |e_2\rangle \langle e_2|. \quad (60)$$

This real space can be thought of a complex space of half the dimension through a complexification map

$$\mathfrak{C}_e = |e_1\rangle \langle e_1| + |ie_2\rangle \langle e_2| \quad (61)$$

giving

$$\mathfrak{C}_e(|a\rangle) = \mathfrak{C}_e(|e_1\rangle a_{11} + |e_2\rangle a_{12}) = |e_1\rangle a_{11} + |ie_1\rangle a_{12} \quad (62)$$

$$= |e_1\rangle a_{11} + |e_1\rangle ia_{12} = |e_1\rangle z_1 = |e_1\rangle R_e(z) \quad (63)$$

This complexification map

$$\mathfrak{C}_e : V_2 \rightarrow V_{2\mathbb{C}}, \quad (64)$$

can also be expressed through the operator complexification, where

$$\mathfrak{C}_e(|e_1\rangle) = \mathfrak{C}_e((1I_2 + 0J_e)|e_1\rangle) = \widehat{\mathfrak{C}}_e(1I_2 + 0J_e)\mathfrak{C}_e(|e_1\rangle) = |e_1\rangle \quad (65)$$

$$\mathfrak{C}_e(|e_2\rangle) = \mathfrak{C}_e((0I_2 + 1J_e)|e_1\rangle) = \widehat{\mathfrak{C}}_e(0I_2 + 1J_e)\mathfrak{C}_e(|e_1\rangle) = i|e_1\rangle \quad (66)$$

giving

$$\mathfrak{C}_e(|a\rangle) = \mathfrak{C}_e(|e_1\rangle a_{11} + |e_2\rangle a_{12}) = \quad (67)$$

$$\mathfrak{C}_e((a_{11}I_2 + a_{12}J_e)|e_1\rangle) = (a_{11} + a_{12}i)|e_1\rangle = z_1|e_1\rangle = |z\rangle, \quad (68)$$

i.e.

$$\mathfrak{C}_e(|a\rangle) = \mathfrak{C}_e(K_e(z_1)|e_1\rangle) = |z\rangle = z_1|e_1\rangle. \quad (69)$$

A complexification and its inverse are given by

$$\mathfrak{C}_e = |e_1\rangle \langle e_1| + |ie_1\rangle \langle e_2| = (|e_1\rangle, |ie_1\rangle) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \langle e_1| \\ \langle e_2| \end{pmatrix} \quad (70)$$

and

$$\mathfrak{C}_e^{-1} = |e_1\rangle \langle e_1|_{V_{2\mathbb{C}}} + |e_2\rangle \langle ie_1|_{V_{2\mathbb{C}}} = |e_1\rangle \langle e_1|_{V_{2\mathbb{C}}} + J_e |e_1\rangle \langle ie_1|_{V_{2\mathbb{C}}} = (|e_1\rangle, |e_2\rangle) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \langle e_1|_{V_{2\mathbb{C}}} \\ \langle ie_1|_{V_{2\mathbb{C}}} \end{pmatrix}, \quad (71)$$

Thus

$$\mathfrak{C}_e = |e_1\rangle (e_1| + i|e_1\rangle (J_e e_1| \quad (72)$$

$$\mathfrak{C}_e(|a\rangle) = (|e_1\rangle (e_1| + i|e_1\rangle (J_e e_1|)|a\rangle = |e_1\rangle a_{11} + i|e_1\rangle a_{12} = (a_{11} + ia_{12})|e_1\rangle = z_1|e_1\rangle = |z\rangle \quad (73)$$

$$\mathfrak{C}_e(|e_1\rangle) = |e_1\rangle \quad (74)$$

$$\mathfrak{C}_e(|e_2\rangle) = \mathfrak{C}_e(|J_e e_1\rangle) = i|e_1\rangle \quad (75)$$

The inverse map

$$\mathfrak{C}_e^{-1} : V_{2e\mathbb{C}} \rightarrow V_2, \quad (76)$$

$$\mathfrak{C}_e^{-1} = |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_2\rangle (ie_1|_{V_{2\mathbb{C}}} = |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + J_e|e_1\rangle (ie_1|_{V_{2\mathbb{C}}} \quad (77)$$

$$\mathfrak{C}_e^{-1}(|z\rangle) = (|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_2\rangle (ie_1|_{V_{2\mathbb{C}}})|z\rangle = |e_1\rangle a_{11} + |e_2\rangle a_{12}. \quad (78)$$

The products of a complexification map and its inverse give the expected results i.e.

$$\mathfrak{C}_e \mathfrak{C}_e^{-1} = (|e_1\rangle (e_1| + i|e_1\rangle (e_2|) (|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_2\rangle (ie_1|_{V_{2\mathbb{C}}}) = I_{V_{2\mathbb{C}}} = |e_1\rangle \langle e_1| = 1 \quad (79)$$

$$\mathfrak{C}_e^{-1} \mathfrak{C}_e = (|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_2\rangle (ie_1|_{V_{2\mathbb{C}}}) (|e_1\rangle (e_1| + i|e_1\rangle (e_2|) = |e_1\rangle (e_1| + |e_2\rangle (e_2| = I_{V_2}. \quad (80)$$

1. Linear Transformation

The relationship between complex linear transformation

$$T_C : V_{2e\mathbb{C}} \rightarrow V_{2e\mathbb{C}} \quad (81)$$

and real linear transformation

$$T_R : V_2 \rightarrow V_2 \quad (82)$$

is given by

$$T_C = \mathfrak{C}_e T_R \mathfrak{C}_e^{-1} = \widehat{\mathfrak{C}_e}(T_R) \quad (83)$$

The action of T_C is given by

$$|w\rangle = T_C |z\rangle = T_C |e_1\rangle R_e(z) = |e_1\rangle R_e(T_C) R_e(z) = |e_1\rangle tz, \quad (84)$$

which can also be expressed as

$$|w\rangle = \mathfrak{C}_e(b) = \mathfrak{C}_e T_R \mathfrak{C}_e^{-1} \mathfrak{C}_e(a) = \mathfrak{C}_e(T_R a). \quad (85)$$

In more detail

$$|w\rangle = T_C |z_1\rangle = T_C |e_1\rangle R_e(z_1) = |e_1\rangle R_e(T_C) R_e(z_1) \quad (86)$$

$$R_e(T_C) R_e(z_1) = (\alpha + i\beta)(x + iy) = (\alpha x - \beta y) + i(\beta x + \alpha y), \quad (87)$$

which can be expressed in terms of the over complete basis $\{|e_1\rangle, |ie_1\rangle\}$ of $V_{2\mathbb{C}}$ using

$$T_C = \begin{pmatrix} |e_1\rangle & |ie_1\rangle \end{pmatrix} \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} (e_1|_{V_{2\mathbb{C}}}) \\ (ie_1|_{V_{2\mathbb{C}}}) \end{pmatrix} \quad (88)$$

$$= (|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}}) R_{ee}(T_C) (|e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}})) \quad (89)$$

$$= |e_1\rangle (e_1|_{V_{2\mathbb{C}}} T_C |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |e_1\rangle (e_1|_{V_{2\mathbb{C}}} T_C |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}} + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}} T_C |e_1\rangle (e_1|_{V_{2\mathbb{C}}} + |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}} T_C |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}})) \quad (90)$$

$$= \alpha |e_1\rangle (e_1|_{V_{2\mathbb{C}}} - \beta |e_1\rangle (ie_1|_{V_{2\mathbb{C}}} + \beta |ie_1\rangle (e_1|_{V_{2\mathbb{C}}} + \alpha |ie_1\rangle (ie_1|_{V_{2\mathbb{C}}})) \quad (91)$$

giving

$$T_C |z\rangle = T_C |e_1, ie_1\rangle R_{ee}(z_1) = |e_1, ie_1\rangle R_{ee}(T_C) R_{ee}(z_1) = T_C |e_1, ie_1\rangle \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} \quad (92)$$

$$= |e_1, ie_1\rangle \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} = |e_1, ie_1\rangle \begin{pmatrix} \alpha a_{11} - \beta a_{12} \\ \beta a_{11} + \alpha a_{12} \end{pmatrix} = |e_1, ie_1\rangle \begin{pmatrix} b_{11} \\ b_{12} \end{pmatrix}. \quad (93)$$

Applying the inverse complexification map gives

$$\widehat{\mathfrak{C}}_e^{-1}(T_C) \mathfrak{C}_e^{-1}(|z_1\rangle) = \widehat{\mathfrak{C}}_e^{-1}(T_C) \mathfrak{C}_e^{-1}(|z_1\rangle) = \widehat{\mathfrak{C}}_e^{-1}(T_C) |\mathfrak{C}_e^{-1}(e_1), \mathfrak{C}_e^{-1}(ie_1)\rangle \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} \quad (94)$$

$$= \widehat{\mathfrak{C}}_e^{-1}(T_C) |e_1, e_2\rangle \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} = |e_1, e_2\rangle \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix} = |e_1, e_2\rangle \begin{pmatrix} \alpha a_{11} - \beta a_{12} \\ \beta a_{11} + \alpha a_{12} \end{pmatrix} = |e_1, e_2\rangle \begin{pmatrix} b_{11} \\ b_{12} \end{pmatrix} \quad (95)$$

Hence

$$R_e(\widehat{\mathfrak{C}}_e^{-1}(t)) = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} = R_{ee}(t). \quad (96)$$

The action of T_R is given by

$$|b\rangle = \mathfrak{C}_e^{-1}(|w\rangle) = \widehat{\mathfrak{C}}_e^{-1}(T_C) \mathfrak{C}_e^{-1}|z\rangle = \mathfrak{C}_e^{-1} T_C \mathfrak{C}_e \mathfrak{C}_e^{-1}|z\rangle, \quad (97)$$

which can also be expressed as

$$|b\rangle = T_R |a\rangle. \quad (98)$$

The representations of T_C and T_R are related by

$$R_e(T_R) = R_e(\mathfrak{C}_e^{-1} T_C \mathfrak{C}_e) = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} = R_e(\mathfrak{C}_e^{-1}) R_{ee}(T_C) R_e(\mathfrak{C}_e) = R_e(\mathfrak{C}_e^{-1}) R_{ee}(t) R_e(\mathfrak{C}_e) \quad (99)$$

$$R_{ee}(T_C) = R_e(\mathfrak{C}_e) R_e(T_R) R_e(\mathfrak{C}_e^{-1}) = R_e(\mathfrak{C}_e) \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} R_e(\mathfrak{C}_e^{-1}), \quad (100)$$

where

$$\mathfrak{C}_e = |e_1\rangle \langle e_1| + |ie_1\rangle \langle e_2| \quad (101)$$

$$\mathfrak{C}_e^{-1} = |e_1\rangle \langle e_1|_{V_{2\mathbb{C}}} + |e_2\rangle \langle ie_1|_{V_{2\mathbb{C}}} = |e_1\rangle \langle e_1|_{V_{2\mathbb{C}}} + J_e |e_1\rangle \langle ie_1|_{V_{2\mathbb{C}}}. \quad (102)$$

$\begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \in \mathcal{B}(\mathbb{R}^2)$ is a representation of $T_C \in \mathcal{B}(\mathbb{C}^1)$. Hence the set of complex linear transformations $\mathcal{B}(\mathbb{C}^1)$ is a real subalgebra of $\mathcal{B}(\mathbb{R}^2)$.

Confirmed by

$$\begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix} \begin{pmatrix} s_R & -s_I \\ s_I & s_R \end{pmatrix} = \begin{pmatrix} t_R s_R - t_I s_I & -(t_R s_I + t_I s_R) \\ (t_R s_I + t_I s_R) & t_R s_R - t_I s_I \end{pmatrix}. \quad (103)$$

Also note

$$\begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -t_I & -t_R \\ t_R & -t_I \end{pmatrix} \quad (104)$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix} = \begin{pmatrix} -t_I & -t_R \\ t_R & -t_I \end{pmatrix} \quad (105)$$

hence

$$\begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix} \quad (106)$$

i.e.

$$\left[\begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] = 0. \quad (107)$$

If

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad (108)$$

then

$$\begin{pmatrix} b & -a \\ d & -c \end{pmatrix} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix} \quad (109)$$

$$\Rightarrow b = -c, a = d \quad (110)$$

hence

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & -c \\ c & a \end{pmatrix}. \quad (111)$$

The subalgebra $\widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^1)) \subset \mathcal{B}(\mathbb{R}^2)$ is thus given by

$$\widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^1)) = \{T_R | [T_R, J_e] = 0; T_R \in \mathcal{B}(\mathbb{R}^2)\} \quad (112)$$

and

$$T_R = t_{11} |e_1\rangle \langle e_1| - t_{12} |e_1\rangle \langle e_2| + t_{12} |e_2\rangle \langle e_1| + t_{11} |e_2\rangle \langle e_2|. \quad (113)$$

a. (Real) $\dim \{V_4\} = 4$ Consider the real space V_4 , where

$$\mathbb{R}\text{-dim} \{V_4\} = 4. \quad (114)$$

The basis is

$$|e_1\rangle = |\mathbf{e}\rangle \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |e_2\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (115)$$

$$|e_3\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, |e_4\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (116)$$

and the complex unit J_e is given by

$$J_e |e_1\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |e_3\rangle \quad (117)$$

$$J_e |e_2\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |e_4\rangle \quad (118)$$

$$J_e |e_3\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |\mathbf{e}\rangle \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = -|e_1\rangle \quad (119)$$

$$J_e |e_4\rangle = |\mathbf{e}\rangle \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |\mathbf{e}\rangle \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} = -|e_2\rangle, \quad (120)$$

i.e.

$$R_e(J_e) = \begin{pmatrix} \mathbb{O} & -\mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O} \end{pmatrix}. \quad (121)$$

Hence

$$J_e = |e_3\rangle \langle e_1| + |e_4\rangle \langle e_2| - |e_1\rangle \langle e_3| - |e_2\rangle \langle e_4|, \quad (122)$$

so corresponds to a real antisymmetric bilinear form, g ,

$$g(a, b) = (g| a \wedge b) = (a| J_e^t b) \quad (123)$$

$$|g\rangle = |e_1 \wedge e_3\rangle + |e_2 \wedge e_4\rangle, \quad (124)$$

⌈

$$(a|J_e^t b) = \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} \quad (125)$$

$$= \begin{pmatrix} a_3 & a_4 & -a_1 & -a_2 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} = a_1 b_3 - a_3 b_1 + a_2 b_4 - a_4 b_2 \quad (126)$$

$$(g|a \wedge b) = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 b_2 - a_2 b_1 \\ a_1 b_3 - a_3 b_1 \\ a_1 b_4 - a_4 b_1 \\ a_2 b_3 - a_3 b_2 \\ a_2 b_4 - a_4 b_2 \\ a_3 b_4 - a_4 b_3 \end{pmatrix} = a_1 b_3 - a_3 b_1 + a_2 b_4 - a_4 b_2 \quad (127)$$

]

while

$$I_4 = |e_1\rangle (e_1| + |e_2\rangle (e_2| + |e_3\rangle (e_3| + |e_4\rangle (e_4|. \quad (128)$$

This real space can be thought of as a complex space of half the dimension through a complexification map

$$\mathfrak{C}_e : V_4 \rightarrow V_{4\mathbb{C}}, \quad (129)$$

given by

$$\mathfrak{C}_e = |e_1\rangle (e_1| + |e_2\rangle (e_2| + |ie_3\rangle (e_3| + |ie_4\rangle (e_4| = \begin{pmatrix} |e_1\rangle & |e_2\rangle & |ie_1\rangle & |ie_2\rangle \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} (e_1| \\ (e_2| \\ (e_3| \\ (e_4| \end{pmatrix}, \quad (130)$$

which has an inverse

$$\mathfrak{C}_e^{-1} : V_{4\mathbb{C}} \rightarrow V_4, \quad (131)$$

given by

$$\mathfrak{C}_e^{-1} = |e_1\rangle (e_1|_{V_{2C}} + |e_2\rangle (e_2|_{V_{2C}} + |e_3\rangle (ie_1|_{V_{2C}} + |e_4\rangle (ie_2|_{V_{2C}} = \begin{pmatrix} |e_1\rangle & |e_2\rangle & |e_3\rangle & |e_4\rangle \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} (e_1|_{V_{2C}} \\ (e_2|_{V_{2C}} \\ (ie_1|_{V_{2C}} \\ (ie_2|_{V_{2C}} \end{pmatrix} \quad (132)$$

The action of $\mathfrak{C}_e$ is

$$\mathfrak{C}_e(|a\rangle) = \mathfrak{C}_e(|e_1\rangle a_{11} + |e_3\rangle a_{12} + |e_2\rangle a_{21} + |e_4\rangle a_{22}) = |e_1\rangle a_{11} + |ie_1\rangle a_{12} + |e_2\rangle a_{21} + |ie_2\rangle a_{22} \quad (133)$$

$$= |e_1\rangle a_{11} + |e_1\rangle ia_{12} + |e_2\rangle a_{21} + |e_2\rangle ia_{22} = |e_1\rangle z_1 + |e_2\rangle z_2 = |e_1, e_2\rangle R_e(z), \quad (134)$$

which can also be expressed as

$$\mathfrak{C}_e(|a\rangle) = \mathfrak{C}_e(|e_1\rangle a_{11} + \mathfrak{C}_e(|e_3\rangle) a_{12} + \mathfrak{C}_e(|e_2\rangle) a_{21} + \mathfrak{C}_e(|e_4\rangle) a_{22}) = \quad (135)$$

$$\mathfrak{C}_e((a_{11}I_4 + a_{12}J_e)|e_1\rangle + (a_{21}I_4 + a_{22}J_e)|e_2\rangle) = (a_{11} + a_{12}i)|e_1\rangle + (a_{21} + a_{22}i)|e_2\rangle = |e_1\rangle z_1 + |e_2\rangle z_2, \quad (136)$$

i.e.

$$\mathfrak{C}_e(|a\rangle) = \mathfrak{C}_e(K_e(z_1)|e_1\rangle + K_e(z_2)|e_2\rangle) = |z_a\rangle = z_1|e_1\rangle + z_2|e_2\rangle = |e_1, e_2\rangle R_e(z_a), \quad (137)$$

where

$$K_e(z_1) = a_{11}R_e(I_4) + a_{12}R_e(J_e) \quad (138)$$

$$K_e(z_2) = a_{21}R_e(I_4) + a_{22}R_e(J_e). \quad (139)$$

Thus one has

$$\mathfrak{C}_e(|a\rangle) = |z\rangle = |z_1e_1 + z_2e_2\rangle = |e_1, e_2\rangle R_e(z) = \mathfrak{C}_e \left(|I_4e_1, J_e e_1, I_4e_2, J_e e_2\rangle \begin{pmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{pmatrix} \right) = \mathfrak{C}_e(|e_1, e_2, e_3, e_4\rangle R_e(z)) \quad (140)$$

where

$$K_e(z_j) = (a_j I_4 + a_{j+2} J_e) \quad (141)$$

is a representation of $\mathbb{C}$ in V_4 .

2. Linear Transformation

The relationship between complex linear transformation

$$T_C : V_{2e\mathbb{C}} \rightarrow V_{2e\mathbb{C}} \quad (142)$$

and real linear transformation

$$T_R : V_2 \rightarrow V_2 \quad (143)$$

is given by

$$T_C = \mathfrak{C}_e T_R \mathfrak{C}_e^{-1} = \widehat{\mathfrak{C}}_e (T_R) \quad (144)$$

The action of T_C is given by

$$|w\rangle = T_C |z\rangle = T_C |e_1, e_2\rangle R_e(z) = |e_1, e_2\rangle R_e(T_C) R_e(z) \quad (145)$$

$$= |e_1, e_2\rangle \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = |e_1, e_2\rangle \begin{pmatrix} x_{11} + iy_{11} & x_{12} + iy_{12} \\ x_{21} + iy_{21} & x_{22} + iy_{22} \end{pmatrix} \begin{pmatrix} a_{11} + ia_{12} \\ a_{21} + ia_{22} \end{pmatrix} \quad (146)$$

$$= |e_1, ie_1, e_2, ie_2\rangle \begin{pmatrix} \mathbb{T}_{11} & \mathbb{T}_{12} \\ \mathbb{T}_{21} & \mathbb{T}_{22} \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{pmatrix} = |e_1, ie_1, e_2, ie_2\rangle \left(\begin{pmatrix} x_{11} & -y_{11} \\ y_{11} & x_{11} \\ x_{21} & -y_{21} \\ y_{21} & x_{21} \end{pmatrix} \begin{pmatrix} x_{12} & -y_{12} \\ y_{12} & x_{12} \\ x_{22} & -y_{22} \\ y_{22} & x_{22} \end{pmatrix} \right) \begin{pmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{pmatrix} \quad (147)$$

$$= |e_1, e_2, ie_1, ie_2\rangle \begin{pmatrix} \mathbb{T}_R & -\mathbb{T}_I \\ \mathbb{T}_I & \mathbb{T}_R \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{pmatrix} = |e_1, e_2, ie_1, ie_2\rangle \left(\begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix} - \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \\ x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \right) \begin{pmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{pmatrix} \quad (148)$$

The subalgebra $\widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2)) \subset \mathcal{B}(\mathbb{R}^4)$ is thus given by

$$\widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2)) = \{T_R | [T_R, J_e] = 0; T_R \in \mathcal{B}(\mathbb{R}^4)\}, \quad (149)$$

and

$$T_R = t_{11} |e_1\rangle \langle e_1| + t_{12} |e_1\rangle \langle e_2| - t_{13} |e_1\rangle \langle e_3| - t_{14} |e_1\rangle \langle e_4| \quad (150)$$

$$t_{21} |e_2\rangle \langle e_1| + t_{22} |e_2\rangle \langle e_2| - t_{23} |e_2\rangle \langle e_3| - t_{24} |e_2\rangle \langle e_4| \quad (151)$$

$$t_{13} |e_3\rangle \langle e_1| + t_{14} |e_3\rangle \langle e_2| + t_{11} |e_3\rangle \langle e_3| + t_{12} |e_3\rangle \langle e_4| \quad (152)$$

$$t_{23} |e_4\rangle \langle e_1| + t_{24} |e_4\rangle \langle e_2| + t_{21} |e_4\rangle \langle e_3| + t_{22} |e_4\rangle \langle e_4|, \quad (153)$$

which shows that $\widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2))$ is an invariant subspace of the one parameter group $\widehat{u}_e$ defined by

$$\widehat{u}_e(x)(T_R) = e^{xJ_e} T_R e^{-xJ_e} = T_R \quad \forall T_R \in \widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2)); x \in \mathbb{R}. \quad (154)$$

The operator J_e belongs to the orthogonal group $O(4, \mathbb{R})$ as

$$(J_e^t = -J_e) \ \& \ (J_e^2 = -I_4) \quad (155)$$

$$\Downarrow \quad (156)$$

$$J_e \in O(\mathbb{R}^4) \quad (157)$$

[

$$(-J_e) J_e = I_4 \Rightarrow J_e^{-1} = (-J_e) = J_e^t \quad (158)$$

]

and it also belongs to the Lie algebra $\mathfrak{o}(4, \mathbb{R})$ as

$$J_e^t = -J_e. \quad (159)$$

Theorem 2 *The set $\widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2)) = \{T_R | [T_R, J_e] = 0; T_R \in \mathcal{B}(\mathbb{R}^4)\}$ is invariant under the transformations $\{\widehat{\mathcal{C}}\}$ where*

$$\widehat{\mathcal{C}}(J_e) = \mathcal{C} J_e \mathcal{C}^t = J_e, \quad (160)$$

which form the adjoint representation of the symplectic group $Sp(4, \mathbb{R})$ that is defined by

$$Sp(4, \mathbb{R}) = \{\mathcal{C} | \mathcal{C} J_e \mathcal{C}^t = J_e; \mathcal{C} \in Gl(4, \mathbb{R}) \subset \mathcal{B}(\mathbb{R}^4)\}. \quad (161)$$

[Note that if $\mathcal{C} \in Sp(4, \mathbb{R})$ then

$$-J_e \mathcal{C} J_e \mathcal{C}^t = I_4 \quad (162)$$

$$(J_e \mathcal{C}^t)^{-1} = -J_e \mathcal{C} \quad (163)$$

$$J_e \mathcal{C}^t J_e \mathcal{C} = -I_4 \quad (164)$$

$$\mathcal{C}^t J_e \mathcal{C} = J_e \quad (165)$$

i.e.

$$\mathcal{C}J_e\mathcal{C}^t = J_e \iff \mathcal{C}^tJ_e\mathcal{C} = J_e. \quad (166)$$

] As

$$[T_R, J_e] = 0 \quad (167)$$

one obtains that

$$\mathcal{C} [T_R, J_e] \mathcal{C}^{-1} = \mathcal{C} T_R^{-1} \mathcal{C} \mathcal{C} J_e^{-1} \mathcal{C} - \mathcal{C} J_e^{-1} \mathcal{C} \mathcal{C} T_R^{-1} \mathcal{C} \quad (168)$$

$$= [\mathcal{C} T_R^{-1} \mathcal{C}, J_e'] = 0. \quad (169)$$

If

$$J'_e = \mathcal{C} J_e^{-1} \mathcal{C} = J_e \quad (170)$$

i.e. $\mathcal{C} J_e^{-1} \mathcal{C} \in \widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2))$ then

$$\mathcal{C}^{-1} = \mathcal{C}^t \quad (171)$$

i.e. $\mathcal{C} \in Sp(4, \mathbb{R})$ and

$$\widehat{\mathcal{C}} : \widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2)) \rightarrow \widehat{\mathfrak{C}}_e^{-1}(\mathcal{B}(\mathbb{C}^2)). \quad (172)$$

If $\mathcal{C} \notin Sp(4, \mathbb{R})$ then

$$\mathcal{C}J_e\mathcal{C}^t = -(\mathcal{C}J_e\mathcal{C}^t)^t \quad (173)$$

and

$$\mathcal{C}J_e\mathcal{C}^t\mathcal{C}J_e\mathcal{C}^t = -I_4 \quad (174)$$

if and only if

$$\mathcal{C}^t\mathcal{C} = \mathcal{C}\mathcal{C}^t = I_4 \quad (175)$$

i.e. $O \in O(4, \mathbb{R})$.

Hence complexifications are related to each by

$$J'_e = OJ_eO^t = \widehat{O}(J_e) \quad (176)$$

$$O \in O(4, \mathbb{R}). \quad (177)$$

The set of unique complexifications $\mathfrak{Com}$ thus corresponds to

$$\Theta \in O(4, \mathbb{R}); \Theta \notin Sp(4, \mathbb{R}) \quad (178)$$

i.e.

$$\mathfrak{Com} = O(4, \mathbb{R}) / (O(4, \mathbb{R}) \cap Sp(4, \mathbb{R})). \quad (179)$$

a. Inner products The inner product in $\mathbb{R}^4$ induces an inner product in $\mathbb{C}^2$ by

$$\langle \omega | z \rangle = \operatorname{Re} \langle \omega | z \rangle + i \operatorname{Im} \langle \omega | z \rangle = r e^{i\theta} \quad (180)$$

$$= \langle \mathfrak{C}_e(a) | \mathfrak{C}_e(b) \rangle = (\mathfrak{C}_e^{-1}(\omega) | \mathfrak{C}_e^{-1}(z)) \quad (181)$$

$$= (a | b) + i (a | J_e b) \quad (182)$$

and then

$$\operatorname{Re} \langle \omega | z \rangle = (a | b) \quad (183)$$

$$\operatorname{Im} \langle \omega | z \rangle = (a | J_e b). \quad (184)$$

In terms of representations

$$R_e(\omega)^\dagger \langle e_1, e_2 | e_1, e_2 \rangle R_e(z) = \begin{pmatrix} \overline{\omega_1} & \overline{\omega_2} \end{pmatrix} \begin{pmatrix} \langle e_1 | e_1 \rangle & \langle e_1 | e_2 \rangle \\ \langle e_2 | e_1 \rangle & \langle e_2 | e_2 \rangle \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \quad (185)$$

$$= \begin{pmatrix} a_{11} - ia_{12} & a_{21} - ia_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b_{11} + ib_{12} \\ b_{21} + ib_{22} \end{pmatrix} \quad (186)$$

$$= \left(\begin{pmatrix} a_{11} & a_{21} \end{pmatrix} - i \begin{pmatrix} a_{12} & a_{22} \end{pmatrix} \right) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \left(\begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix} + i \begin{pmatrix} b_{12} \\ b_{22} \end{pmatrix} \right) \quad (187)$$

$$= \begin{pmatrix} a_{11} & a_{21} \end{pmatrix} \begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix} + \begin{pmatrix} a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} b_{12} \\ b_{22} \end{pmatrix} + i \left(\begin{pmatrix} a_{11} & a_{21} \end{pmatrix} \begin{pmatrix} b_{12} \\ b_{22} \end{pmatrix} - \begin{pmatrix} a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix} \right) \quad (188)$$

$$= \begin{pmatrix} a_{11} & a_{21} & a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix} \begin{pmatrix} b_{11} \\ b_{21} \\ b_{12} \\ b_{22} \end{pmatrix} \quad (189)$$

$$+ i \begin{pmatrix} a_{11} & a_{21} & a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix} \begin{pmatrix} b_{11} \\ b_{21} \\ b_{12} \\ b_{22} \end{pmatrix} \quad (190)$$

$$= R_e(a)^t S_e R_e(b) + i R_e(a)^t J_e R_e(b). \quad (191)$$

If $T_R \in O(\mathbb{R}^4) \cap Sp(\mathbb{R}^4)$ i.e.

$$T_R^t T_R = I_{\mathbb{R}^4} \quad (192)$$

$$T_R^t J_e T_R = J_e \quad (193)$$

then

$$[T_R, J_e] = 0 \quad (194)$$

and $T_R \in U(\mathbb{C}^2)$ i.e.

$$\widehat{\mathfrak{C}}_e(T_R)^\dagger \widehat{\mathfrak{C}}_e(T_R) = I_{\mathbb{C}^2} \quad (195)$$

as

$$\langle T_{\mathbb{C}}\omega | T_{\mathbb{C}}z \rangle = \text{Re} \langle T_{\mathbb{C}}\omega | T_{\mathbb{C}}z \rangle + i \text{Im} \langle T_{\mathbb{C}}\omega | T_{\mathbb{C}}z \rangle = r e^{i\theta} \quad (196)$$

$$= (T_{\mathbb{R}}a | T_{\mathbb{R}}b) + i (T_{\mathbb{R}}a | J_e T_{\mathbb{R}}b) \quad (197)$$

$$= (a | b) + i (a | J_e b). \quad (198)$$

This shows that

$$\mathfrak{Com} = O(4, \mathbb{R}) / U(2, \mathbb{C}). \quad (199)$$

The different complex algebras represented in $\mathcal{B}(\mathbb{R}^4)$ are thus connected by

$$\widehat{\Theta} : \widehat{\mathfrak{C}}_{e_1}^{-1}(\mathcal{B}(\mathbb{C}^2)) \rightarrow \widehat{\mathfrak{C}}_{e_2}^{-1}(\mathcal{B}(\mathbb{C}^2)); \widehat{\Theta} \in \mathfrak{Com}. \quad (200)$$

C. Questions

- $J_{e_1} \neq J_{e_2}$

$$[T_R, J_{e_1}] = 0 \Rightarrow [T_R, J_{e_2}] \neq 0 \quad (201)$$

$$T_R J_{e_1} - J_{e_1} T_R = 0 \Leftrightarrow T_R = J_{e_1} T_R J_{e_1}^t \quad (202)$$

$$T_R J_{e_2} - J_{e_2} T_R = 0 \Leftrightarrow T_R = J_{e_2} T_R J_{e_2}^t \quad (203)$$

$$\Rightarrow J_{e_2} T_R J_{e_2}^t = J_{e_1} T_R J_{e_1}^t \quad (204)$$

$$\Rightarrow T_R = J_{e_2}^t J_{e_1} T_R J_{e_1}^t J_{e_2} \quad (205)$$

$$\Rightarrow J_{e_1}^t J_{e_2} T_R = T_R J_{e_1}^t J_{e_2} \quad (206)$$

$$J_{e_2}^t J_{e_1} = J_{e_2} J_{e_1}^t \quad (207)$$

$$(J_{e_2}^t J_{e_1})^t = J_{e_1}^t J_{e_2} = J_{e_1} J_{e_2}^t \quad (208)$$

•

$$\exists J_e \forall T_R \in \mathcal{B}(\mathbb{R}^4) \quad (209)$$

•

$$\mathcal{B}(\mathbb{R}^4) \quad (210)$$

$$\widehat{\mathfrak{C}}_{e_1} \swarrow \cdots \searrow \widehat{\mathfrak{C}}_{e_2} \quad (211)$$

$$\mathcal{B}(\mathbb{C}^2)_{e_1} \xrightarrow{u_{12}} \mathcal{B}(\mathbb{C}^2)_{e_2} \quad (212)$$

Complex units are related to each other by

$$J_{e_2} = \widehat{O_{12}}(J_{e_1}) = O_{12} J_{e_1} O_{12}^t; O_{12} \in O(4, \mathbb{R}), \quad (213)$$

which clearly shows that if $O_{12} \in O(4, \mathbb{R}) \cap Sp(4, \mathbb{R})$ then $J_{e_2} = J_{e_1}$.

$$T_R = t_{11} |e_1\rangle \langle e_1| + t_{12} |e_1\rangle \langle e_2| - t_{13} |e_1\rangle \langle e_3| - t_{14} |e_1\rangle \langle e_4| \quad (214)$$

$$t_{21} |e_2\rangle \langle e_1| + t_{22} |e_2\rangle \langle e_2| - t_{23} |e_2\rangle \langle e_3| - t_{24} |e_2\rangle \langle e_4| \quad (215)$$

$$t_{13} |e_3\rangle \langle e_1| + t_{14} |e_3\rangle \langle e_2| + t_{11} |e_3\rangle \langle e_3| + t_{12} |e_3\rangle \langle e_4| \quad (216)$$

$$t_{23} |e_4\rangle \langle e_1| + t_{24} |e_4\rangle \langle e_2| + t_{21} |e_4\rangle \langle e_3| + t_{22} |e_4\rangle \langle e_4|, \quad (217)$$

$$T_R = \begin{pmatrix} |\mathbf{e}_R\rangle & |\mathbf{e}_I\rangle \end{pmatrix} \begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix} \begin{pmatrix} \langle \mathbf{e}_R| \\ \langle \mathbf{e}_I| \end{pmatrix} \quad (218)$$

$$T_R = s_{11} |f_1\rangle \langle f_1| + s_{12} |f_1\rangle \langle f_2| - s_{13} |f_1\rangle \langle f_3| - s_{14} |f_1\rangle \langle f_4| \quad (219)$$

$$s_{21} |f_2\rangle \langle f_1| + s_{22} |f_2\rangle \langle f_2| - s_{23} |f_2\rangle \langle f_3| - s_{24} |f_2\rangle \langle f_4| \quad (220)$$

$$s_{13} |f_3\rangle \langle f_1| + s_{14} |f_3\rangle \langle f_2| + s_{11} |f_3\rangle \langle f_3| + s_{12} |f_3\rangle \langle f_4| \quad (221)$$

$$s_{23} |f_4\rangle \langle f_1| + s_{24} |f_4\rangle \langle f_2| + s_{21} |f_4\rangle \langle f_3| + s_{22} |f_4\rangle \langle f_4|, \quad (222)$$

$$T_R = \begin{pmatrix} |\mathbf{f}_R\rangle & |\mathbf{f}_I\rangle \end{pmatrix} \begin{pmatrix} s_R & -s_I \\ s_I & s_R \end{pmatrix} \begin{pmatrix} \langle \mathbf{f}_R| \\ \langle \mathbf{f}_I| \end{pmatrix} = \begin{pmatrix} |\mathbf{e}_R\rangle & |\mathbf{e}_I\rangle \end{pmatrix} R_e(O_{12}^t) \begin{pmatrix} s_R & -s_I \\ s_I & s_R \end{pmatrix} R_e(O_{12}) \begin{pmatrix} \langle \mathbf{e}_R| \\ \langle \mathbf{e}_I| \end{pmatrix} \quad (223)$$

$$\begin{pmatrix} t_R & -t_I \\ t_I & t_R \end{pmatrix} = R_e(O_{12}^t) \begin{pmatrix} s_R & -s_I \\ s_I & s_R \end{pmatrix} R_e(O_{12}) \quad (224)$$

Hence

$$T_R J_{e_2} - J_{e_2} T_R = 0 \quad (225)$$

$$\Rightarrow T_R O_{12} J_{e_1} O_{12}^t - O_{12} J_{e_1} O_{12}^t T_R = 0 \quad (226)$$

$$\Rightarrow O_{12}^t T_R O_{12} J_{e_1} - J_{e_1} O_{12}^t T_R O_{12} = 0 \quad (227)$$

i.e.

$$O_{12}^t T_R O_{12} \in \widehat{\mathfrak{C}}_{e_1}^{-1}(\mathcal{B}(\mathbb{C}^2)). \quad (228)$$

So if

$$O_{12}^t T_R O_{12} \text{ and } T_R \in \widehat{\mathfrak{C}}_{e_1}^{-1}(\mathcal{B}(\mathbb{C}^2)) \quad (229)$$

then

$$T_R J_{e_1} - J_{e_1} T_R = 0 \Rightarrow T_R = J_{e_1}^t T_R J_{e_1} \quad (230)$$

$$O_{12}^t T_R O_{12} J_{e_1} - J_{e_1} O_{12}^t T_R O_{12} = 0 \Rightarrow T_R = O_{12} J_{e_1} O_{12}^t T_R O_{12} J_{e_1}^t O_{12}^t. \quad (231)$$

⌈

$$T_R J_{e_2} - J_{e_2} T_R = 0 \Leftrightarrow J_{e_2}^t T_R^t - T_R^t J_{e_2}^t = T_R^t J_{e_2} - J_{e_2} T_R^t = [T_R^t, J_{e_2}] \quad (232)$$

⌋

D. Groups

Let

$$Q \begin{pmatrix} |\mathbf{e}_R\rangle & |\mathbf{e}_I\rangle \end{pmatrix} = \begin{pmatrix} |\mathbf{e}_R\rangle & |\mathbf{e}_I\rangle \end{pmatrix} \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} \quad (233)$$

then

$$Q \in O(4, \mathbb{R}) \Leftrightarrow \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix}^t = \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} \begin{pmatrix} Q_{RR}^t & Q_{IR}^t \\ Q_{RI}^t & Q_{II}^t \end{pmatrix} \quad (234)$$

$$= \begin{pmatrix} \mathbb{I}_2 & \mathbb{O} \\ \mathbb{O} & \mathbb{I}_2 \end{pmatrix} = \begin{pmatrix} Q_{RR}Q_{RR}^t + Q_{RI}Q_{RI}^t & Q_{RR}Q_{IR}^t + Q_{RI}Q_{II}^t \\ Q_{IR}Q_{RR}^t + Q_{II}Q_{RI}^t & Q_{IR}Q_{IR}^t + Q_{II}Q_{II}^t \end{pmatrix} \quad (235)$$

$$Q_{RR}Q_{RR}^t + Q_{RI}Q_{RI}^t = \mathbb{I}_2 \quad (236)$$

$$Q_{IR}Q_{IR}^t + Q_{II}Q_{II}^t = \mathbb{I}_2 \quad (237)$$

$$Q_{IR}Q_{RR}^t + Q_{II}Q_{RI}^t = \mathbb{O} \quad (238)$$

$$Q_{RR}Q_{IR}^t + Q_{RI}Q_{II}^t = \mathbb{O} \quad (239)$$

and

$$Q \in Sp(4, \mathbb{R}) \Leftrightarrow \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} \begin{pmatrix} \mathbb{O} & -\mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O} \end{pmatrix} \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix}^t \quad (240)$$

$$= \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} \begin{pmatrix} \mathbb{O} & -\mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O} \end{pmatrix} \begin{pmatrix} Q_{RR}^t & Q_{IR}^t \\ Q_{RI}^t & Q_{II}^t \end{pmatrix} \quad (241)$$

$$= \begin{pmatrix} \mathbb{O} & -\mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O} \end{pmatrix} = \begin{pmatrix} Q_{RI} & -Q_{RR} \\ Q_{II} & -Q_{IR} \end{pmatrix} \begin{pmatrix} Q_{RR}^t & Q_{IR}^t \\ Q_{RI}^t & Q_{II}^t \end{pmatrix} = \begin{pmatrix} Q_{RI}Q_{RR}^t - Q_{RR}Q_{RI}^t & Q_{RI}Q_{IR}^t - Q_{RR}Q_{II}^t \\ Q_{II}Q_{RR}^t - Q_{IR}Q_{RI}^t & Q_{II}Q_{IR}^t - Q_{IR}Q_{II}^t \end{pmatrix} \quad (242)$$

$$Q_{RI}Q_{RR}^t - Q_{RR}Q_{RI}^t = \mathbb{O} \quad (243)$$

$$Q_{II}Q_{IR}^t - Q_{IR}Q_{II}^t = \mathbb{O} \quad (244)$$

$$Q_{II}Q_{RR}^t - Q_{IR}Q_{RI}^t = \mathbb{I}_2 \quad (245)$$

$$Q_{RI}Q_{IR}^t - Q_{RR}Q_{II}^t = -\mathbb{I}_2 \quad (246)$$

If $Q \in Sp(4, \mathbb{R}) \cap O(4, \mathbb{R})$ then

- from $Q \in O(4, \mathbb{R})$

$$Q_{RR}Q_{RR}^t + Q_{RI}Q_{RI}^t = \mathbb{I}_2 \quad (247)$$

$$Q_{IR}Q_{IR}^t + Q_{II}Q_{II}^t = \mathbb{I}_2 \quad (248)$$

$$Q_{RI}Q_{RR}^t = Q_{RR}Q_{RI}^t \Rightarrow Q_{RI} = Q_{RR}Q_{RI}^tQ_{II} \quad (249)$$

$$Q_{II}Q_{IR}^t = Q_{IR}Q_{II}^t \Rightarrow Q_{IR}^t = Q_{RR}^tQ_{IR}Q_{II}^t \quad (250)$$

- from $Q \in Sp(4, \mathbb{R})$

$$Q_{II}Q_{RR}^t - Q_{IR}Q_{RI}^t = Q_{RR}Q_{II}^t - Q_{RI}Q_{IR}^t = \mathbb{I}_2 \quad (251)$$

$$Q_{IR}Q_{RR}^t = -Q_{II}Q_{RI}^t \Rightarrow Q_{IR} = -Q_{II}Q_{RI}^tQ_{II} \quad (252)$$

$$Q_{RR}Q_{IR}^t = -Q_{RI}Q_{II}^t \Rightarrow Q_{IR}^t = -Q_{II}^tQ_{RI}Q_{II}^t \quad (253)$$

- from $Q \in Sp(4, \mathbb{R}) \cap O(4, \mathbb{R})$

$$\Rightarrow Q_{II}Q_{RR}^t - (\mathbb{I}_2 - Q_{II}Q_{RR}^t) = \mathbb{I}_2 \quad (254)$$

$$\Rightarrow Q_{II}Q_{RR}^t = Q_{RR}Q_{II}^t = Q_{II}^tQ_{RR} = Q_{RR}^tQ_{II} = \mathbb{I}_2 \quad (255)$$

i.e.

$$Q_{II}^{-1} = Q_{RR}^t \quad (256)$$

$$Q_{RR}^{-1} = Q_{II}^t \quad (257)$$

- from $Q \in Sp(4, \mathbb{R}) \cap O(4, \mathbb{R})$

$$\mathbb{I}_2 = Q_{RR}Q_{II}^t - Q_{RI}Q_{IR}^t \quad (258)$$

$$\mathbb{I}_2 = Q_{RR}Q_{II}^t \quad (259)$$

$$\Rightarrow Q_{RI}Q_{IR}^t = \mathbb{O} = -(Q_{RI}Q_{II}^t)^2 \Rightarrow Q_{RI} = 0 \quad (260)$$

hence

$$\mathbb{I}_2 = Q_{RR}Q_{II}^t \quad (261)$$

and

$$Q_{RR}Q_{RR}^t = \mathbb{I}_2 \quad (262)$$

$$Q_{IR}Q_{IR}^t = \mathbb{I}_2 \quad (263)$$

thus

$$Q_{II}^{-1} = Q_{RR}^t = Q_{II}^t \Rightarrow Q_{RR} = Q_{II}. \quad (264)$$

Hence

$$Q = \begin{pmatrix} Q_2 & \mathbb{O} \\ \mathbb{O} & Q_2 \end{pmatrix} \quad (265)$$

and

$$Q \in O(2, 2, \mathbb{R}). \quad (266)$$

If

$$Q \in O(2, 2, \mathbb{R}) \quad (267)$$

then

$$\begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} \begin{pmatrix} \mathbb{I}_2 & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_2 \end{pmatrix} \begin{pmatrix} Q_{RR}^t & Q_{IR}^t \\ Q_{RI}^t & Q_{II}^t \end{pmatrix} \quad (268)$$

$$= \begin{pmatrix} Q_{RR} & -Q_{RI} \\ Q_{IR} & -Q_{II} \end{pmatrix} \begin{pmatrix} Q_{RR}^t & Q_{IR}^t \\ Q_{RI}^t & Q_{II}^t \end{pmatrix} = \quad (269)$$

$$\begin{pmatrix} Q_{RR}Q_{RR}^t - Q_{RI}Q_{RI}^t & Q_{RR}Q_{IR}^t - Q_{RI}Q_{II}^t \\ Q_{IR}Q_{RR}^t - Q_{II}Q_{RI}^t & Q_{IR}Q_{IR}^t - Q_{II}Q_{II}^t \end{pmatrix} = \begin{pmatrix} \mathbb{I}_2 & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_2 \end{pmatrix} \quad (270)$$

$$Q_{RR}Q_{RR}^t - Q_{RI}Q_{RI}^t = \mathbb{I}_2 \quad (271)$$

$$Q_{RR}Q_{IR}^t - Q_{RI}Q_{II}^t = \mathbb{O} \quad (272)$$

$$Q_{IR}Q_{IR}^t - Q_{II}Q_{II}^t = \mathbb{I}_2 \quad (273)$$

We have from the precedings that

$$Q \in O(4, \mathbb{R}) \cap Sp(4, \mathbb{R}) \Rightarrow Q \in O(2, 2, \mathbb{R}) \quad (274)$$

but

$$Q \in O(2, 2, \mathbb{R}) \not\Rightarrow Q \in O(4, \mathbb{R}) \cap Sp(4, \mathbb{R}). \quad (275)$$

The various complex units are given by

$$OJ_e O^t = O|e_3\rangle(e_1|O^t + O|e_4\rangle(e_2|O^t - O|e_1\rangle(e_3|O^t - O|e_2\rangle(e_4|O^t \quad (276)$$

⌈

$$(iJ_e)^\dagger = -iJ_e^t = iJ_e \quad (277)$$

$$(iJ_e)^\dagger (iJ_e) = iJ_e iJ_e = I \quad (278)$$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \lambda \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -ib \\ ia \end{pmatrix} = \begin{pmatrix} -iia \\ ia \end{pmatrix} = \begin{pmatrix} a \\ ia \end{pmatrix} = \begin{pmatrix} a \\ ia \end{pmatrix} \quad (279)$$

The solutions for λ come from the secular equation

$$\begin{vmatrix} \lambda & -i \\ i & \lambda \end{vmatrix} = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \quad (280)$$

which for $\lambda = 1$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -ib \\ ia \end{pmatrix} = \begin{pmatrix} -iia \\ ia \end{pmatrix} = \begin{pmatrix} a \\ ia \end{pmatrix} = \begin{pmatrix} a \\ ia \end{pmatrix} \quad (281)$$

gives the eigen vector

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (282)$$

and for $\lambda = -1$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ib \\ -ia \end{pmatrix} = \begin{pmatrix} -iia \\ ia \end{pmatrix} = \begin{pmatrix} a \\ ia \end{pmatrix} = - \begin{pmatrix} a \\ ia \end{pmatrix} \quad (283)$$

gives the eigen vector

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \quad (284)$$

which is displayed by

$$\begin{aligned} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \\ \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} &= \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ i \end{pmatrix} = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \end{aligned}$$

This is confirmed by expanding

$$iJ_e = |v_1\rangle \langle v_1| - |v_2\rangle \langle v_2| \quad (285)$$

to obtain

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ i \end{pmatrix} \begin{pmatrix} 1 & -i \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix} \begin{pmatrix} 1 & i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}. \quad (286)$$

corresponding to

$$J_e = -i \{ |v_1\rangle \langle v_1| - |v_2\rangle \langle v_2| \} \quad (287)$$

hence

$$-i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -i \\ 1 \end{pmatrix} \begin{pmatrix} 1 & -i \end{pmatrix} + \frac{1}{2} \begin{pmatrix} i \\ 1 \end{pmatrix} \begin{pmatrix} 1 & i \end{pmatrix}. \quad (288)$$

corresponding to

$$iJ_{4e} = |v_1\rangle \langle v_1| + |v_2\rangle \langle v_2| - (|v_3\rangle \langle v_3| + |v_4\rangle \langle v_4|) = P_1 - P_2 \quad (289)$$

]

Define

$$P_R = |e_1\rangle \langle e_1| + |e_2\rangle \langle e_2| \quad (290)$$

$$P_I = |e_3\rangle \langle e_3| + |e_4\rangle \langle e_4| \quad (291)$$

then

$$J_{4e} = |e_3\rangle \langle e_1| + |e_4\rangle \langle e_2| - |e_1\rangle \langle e_3| - |e_2\rangle \langle e_4| \quad (292)$$

and

$$P_R J_{4e} P_I = -|e_1\rangle \langle e_3| - |e_2\rangle \langle e_4| \quad (293)$$

$$P_I J_{4e} P_R = |e_3\rangle \langle e_1| + |e_4\rangle \langle e_2|. \quad (294)$$

Ffrom the preceding $Q \in Sp(4, \mathbb{R}) \cap O(4, \mathbb{R})$ if

$$[Q, P_R] = [Q, P_I] = 0 \quad (295)$$

I. COMPLEXIFICATION AS REPRESENTATIONS OF $\mathbb{C}$

Consider a real even dimensional Hilbert space $\mathcal{V}$

$$\dim \{\mathcal{V}\} = 2s \quad (296)$$

and the set of bounded linear transformations $\mathcal{B}(\mathcal{V})$.

An orthogonal complexification is given by a complex unit $J_e \in \mathcal{B}(\mathcal{V})$ that has the properties

$$J_e^t = -J_e \text{ \& } J_e^2 = -I_V \quad (297)$$

$$\Downarrow \quad (298)$$

$$J_e \in O(\mathbb{R}^4) \quad (299)$$

as

$$J_e^2 = -I_{\mathcal{V}} \quad (300)$$

$$J_e^t = -J_e \quad (301)$$

Where

$$J_e = \sum_{1 \leq j \leq s} \{|e_j\rangle (e_{j+s}| - |e_{j+s}\rangle (e_j|\} . \quad (302)$$

The identity belonging to $\mathcal{B}(\mathcal{V})$ can be expressed as

$$I_{\mathcal{V}} = \sum_{1 \leq j \leq 2s} |e_j\rangle (e_j| = P_e + P_{\bar{e}}, \quad (303)$$

where

$$P_e = \sum_{1 \leq j \leq s} |e_j\rangle (e_j| \quad (304)$$

$$P_{\bar{e}} = \sum_{s+1 \leq j \leq 2s} |e_j\rangle (e_j|. \quad (305)$$

The complex numbers $\mathbb{C}$ can be represented in $\mathcal{B}(\mathcal{V})$ by

$$K_e(z) = \text{Re } z I_e + \text{Im } z J_e = I_e \text{Re } z + J_e \text{Im } z; z \in \mathbb{C} \quad (306)$$

[

$$z_1 z_2 = (\text{Re } z_1 + i \text{Im } z_1) (\text{Re } z_2 + i \text{Im } z_2) = (\text{Re } z_1 \text{Re } z_2 - \text{Im } z_1 \text{Im } z_2) + i (\text{Im } z_1 \text{Re } z_2 + \text{Re } z_1 \text{Im } z_2) \quad (307)$$

$$K_e(z_1 z_2) = (\text{Re } z_1 \text{Re } z_2 - \text{Im } z_1 \text{Im } z_2) I_e + (\text{Im } z_1 \text{Re } z_2 + \text{Re } z_1 \text{Im } z_2) J_e \quad (308)$$

$$K_e(z_1) K_e(z_2) = (\text{Re } z_1 I_e + \text{Im } z_1 J_e) (\text{Re } z_2 I_e + \text{Im } z_2 J_e) = (\text{Re } z_1 \text{Re } z_2 - \text{Im } z_1 \text{Im } z_2) I_e + (\text{Im } z_1 \text{Re } z_2 + \text{Re } z_1 \text{Im } z_2) J_e \quad (309)$$

$$K_e(z_1 z_2) = K_e(z_1) K_e(z_2) \quad (310)$$

]

inducing the complexification

$$|a\rangle = |e_1, e_2, e_3, e_4\rangle \begin{pmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{pmatrix} = |e_1, e_2, J_e e_1, J_e e_2\rangle \begin{pmatrix} a_{11} \\ a_{21} \\ a_{12} \\ a_{22} \end{pmatrix} \quad (311)$$

$$= |e_1\rangle a_{11} + |e_2\rangle a_{21} + J_e |e_1\rangle a_{12} + J_e |e_2\rangle a_{22} \quad (312)$$

$$= a_{11} I_{2s} |e_1\rangle + a_{21} I_{2s} |e_2\rangle + a_{12} J_e |e_1\rangle + a_{22} J_e |e_2\rangle \quad (313)$$

$$= (a_{11} I_{2s} + a_{12} J_e) |e_1\rangle + (a_{21} I_{2s} + a_{22} J_e) |e_2\rangle \quad (314)$$

$$= K_e(z_1) |e_1\rangle + K_e(z_2) |e_2\rangle \quad (315)$$

$$= |K_e(z_1), K_e(z_2)\rangle \begin{pmatrix} |e_1\rangle \\ |e_2\rangle \end{pmatrix}, \quad (316)$$

where

$$K_e(z_1) = a_{11} I_{2s} + a_{12} J_e = \text{Re } z_1 I_e + \text{Im } z_1 J_e = |I_{2s}, J_e\rangle \begin{pmatrix} \text{Re } z_1 \\ \text{Im } z_1 \end{pmatrix} \quad (317)$$

$$K_e(z_2) = a_{21} I_{2s} + a_{22} J_e = \text{Re } z_2 I_e + \text{Im } z_2 J_e = |I_{2s}, J_e\rangle \begin{pmatrix} \text{Re } z_2 \\ \text{Im } z_2 \end{pmatrix}. \quad (318)$$

In order to see that this a complexification one expands

$$K_e(z_1) |e_1\rangle + K_e(z_2) |e_2\rangle = |I_{2s}, J_e\rangle \begin{pmatrix} \text{Re } z_1 \\ \text{Im } z_1 \end{pmatrix} |e_1\rangle + |I_{2s}, J_e\rangle \begin{pmatrix} \text{Re } z_2 \\ \text{Im } z_2 \end{pmatrix} |e_2\rangle \quad (319)$$

$$= |I_{2s}, J_e\rangle \left\{ \begin{pmatrix} \text{Re } z_1 \\ \text{Im } z_1 \end{pmatrix} |e_1\rangle + \begin{pmatrix} \text{Re } z_2 \\ \text{Im } z_2 \end{pmatrix} |e_2\rangle \right\} \quad (320)$$

$$= |I_{2s}, J_e\rangle \left\{ \left| \begin{pmatrix} \text{Re } z_1 \\ \text{Im } z_1 \end{pmatrix}, \begin{pmatrix} \text{Re } z_2 \\ \text{Im } z_2 \end{pmatrix} \right\rangle \begin{pmatrix} |e_1\rangle \\ |e_2\rangle \end{pmatrix} \right\} \quad (321)$$

$$= |I_{2s}, J_e\rangle \left\{ |z_1, z_2\rangle \begin{pmatrix} |e_1\rangle \\ |e_2\rangle \end{pmatrix} \right\}. \quad (322)$$

Also

$$K_e(z_1)|e_1\rangle + K_e(z_2)|e_2\rangle = |I_{2s}, J_e\rangle \left\{ ||e_1\rangle, |e_2\rangle \right) \left(\begin{pmatrix} \operatorname{Re} z_1 \\ \operatorname{Im} z_1 \\ \operatorname{Re} z_2 \\ \operatorname{Im} z_2 \end{pmatrix} \right) \right\} \quad (323)$$

$$= |I_{2s}, J_e\rangle \left\{ ||e_1\rangle, |e_2\rangle \right) \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right\}, \quad (324)$$

where

$$z \leftrightarrow \begin{pmatrix} \operatorname{Re} z \\ \operatorname{Im} z \end{pmatrix} \leftrightarrow (1, i) \begin{pmatrix} \operatorname{Re} z \\ \operatorname{Im} z \end{pmatrix} = 1(\operatorname{Re} z) + i(\operatorname{Im} z) \in \mathcal{B}(\mathbb{R}^2) \quad (325)$$

$$1, i \in \mathcal{B}(\mathbb{R}^2). \quad (326)$$

II. REPRESENTATION OF $\mathcal{V}$ IN $\mathcal{B}(\mathcal{V})$

The space $\mathcal{V}$ can be represented in $\mathcal{B}(\mathcal{V})$ by the linear injection

$$K_e : \mathcal{V} \hookrightarrow \mathcal{B}(\mathcal{V}) \quad (327)$$

$$|e_j\rangle \rightarrow |e_j\rangle(e_j| \quad (328)$$

$$K_e(a) = \sum_{1 \leq j \leq 2s} |e_j\rangle(e_j| R_e(a)_j \quad (329)$$

$$|a\rangle = \sum_{1 \leq j \leq 2s} |e_j\rangle R_e(a)_j. \quad (330)$$

K_e is a Hilbert space isomorphism as

$$K_e : \mathcal{V} \hookrightarrow K_e(\mathcal{V}) \subset \mathcal{B}(\mathcal{V}) \quad (331)$$

$$(a|b) = \operatorname{Tr} \{ K_e(a)^t K_e(b) \}. \quad (332)$$

$$K_e(\mathcal{V}_{\mathbb{C}}) = \left\{ (|e_1\rangle(e_1|, \dots, |e_{2s}\rangle(e_{2s}|) \begin{pmatrix} a_1 \\ \vdots \\ a_{2s} \end{pmatrix} \right\} \quad (333)$$

$$= \left\{ (|e_1\rangle\langle e_1|, \dots, |e_s\rangle\langle e_s|, |e_{s+1}\rangle\langle e_{s+1}|, \dots, |e_{2s}\rangle\langle e_{2s}|) \begin{pmatrix} a_1 \\ \vdots \\ a_s \\ a_{s+1} \\ \vdots \\ a_{2s} \end{pmatrix} \right\} \quad (334)$$

$$= (|e_1\rangle\langle e_1|, \dots, |e_s\rangle\langle e_s|) \begin{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_s \end{pmatrix} \end{pmatrix} \quad (335)$$

$$+ ((J_e |e_1\rangle\langle e_1| J_e^t, \dots, J_e |e_s\rangle\langle e_s| J_e^t) \begin{pmatrix} \begin{pmatrix} a_{s+1} \\ \vdots \\ a_{2s} \end{pmatrix} \end{pmatrix}) \quad (336)$$

$$= \left\{ (|e_1\rangle\langle e_1|, \dots, |e_s\rangle\langle e_s|, J_e |e_1\rangle\langle e_1| J_e^t, \dots, J_e |e_s\rangle\langle e_s| J_e^t) \begin{pmatrix} a_1 \\ \vdots \\ a_s \\ a_{s+1} \\ \vdots \\ a_{2s} \end{pmatrix} \right\} \quad (337)$$

$$= \left\{ (|e_1\rangle\langle e_1|, \dots, |e_s\rangle\langle e_s|, J_e |e_1\rangle\langle e_1| J_e^t, \dots, J_e |e_s\rangle\langle e_s| J_e^t) \begin{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_s \end{pmatrix} \\ \begin{pmatrix} a_{s+1} \\ \vdots \\ a_{2s} \end{pmatrix} \end{pmatrix} \right\} \quad (338)$$

$$= \left\{ \left((|e_1\rangle\langle e_1|, \dots, |e_s\rangle\langle e_s|), \left(\widehat{J}_e(|e_1\rangle\langle e_1|), \dots, \widehat{J}_e(|e_s\rangle\langle e_s|) \right) \right) \left(\begin{pmatrix} a_1 \\ \vdots \\ a_s \\ a_{s+1} \\ \vdots \\ a_{2s} \end{pmatrix} \right) \right\}. \quad (339)$$

$$J_e |e_k\rangle\langle e_k| J_e^t = |e_{k+s}\rangle\langle e_{k+s}| \quad (340)$$

$$J_e |e_{k+s}\rangle\langle e_{k+s}| J_e^t = |e_k\rangle\langle e_k| \quad (341)$$

$$\begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} \begin{pmatrix} \mathbb{I}_2 & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_2 \end{pmatrix} = \begin{pmatrix} \mathbb{I}_2 & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_2 \end{pmatrix} \begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix}^t \quad (342)$$

$$\Updownarrow \quad (343)$$

$$\begin{pmatrix} Q_{RR} & -Q_{RI} \\ Q_{IR} & -Q_{II} \end{pmatrix} = \begin{pmatrix} Q_{RR}^t & -Q_{IR}^t \\ Q_{RI}^t & -Q_{II}^t \end{pmatrix} \quad (344)$$

$$\Updownarrow \quad (345)$$

$$\begin{pmatrix} Q_{RR} & Q_{RI} \\ Q_{IR} & Q_{II} \end{pmatrix} = \begin{pmatrix} Q_{RR} & \mathbb{O} \\ \mathbb{O} & Q_{II} \end{pmatrix} \quad (346)$$

$$\begin{pmatrix} Q_{RR} & \mathbb{O} \\ \mathbb{O} & Q_{II} \end{pmatrix} \begin{pmatrix} \mathbb{O} & -\mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O} \end{pmatrix} \begin{pmatrix} Q_{RR} & \mathbb{O} \\ \mathbb{O} & Q_{II} \end{pmatrix}^t = \begin{pmatrix} \mathbb{O} & -\mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O} \end{pmatrix} \quad (347)$$

$$\begin{pmatrix} \mathbb{O} & -Q_{RR} \\ Q_{II} & \mathbb{O} \end{pmatrix} \begin{pmatrix} Q_{RR}^t & \mathbb{O} \\ \mathbb{O} & Q_{II}^t \end{pmatrix} = \begin{pmatrix} \mathbb{O} & -Q_{RR}Q_{II}^t \\ Q_{II}Q_{RR}^t & \mathbb{O} \end{pmatrix} = \begin{pmatrix} \mathbb{O} & -\mathbb{I}_2 \\ \mathbb{I}_2 & \mathbb{O} \end{pmatrix} \quad (348)$$